

1 Backend is not one thing — it's an ecosystem

A real backend system is made of:

Layer	Purpose
Client	Browser, Mobile App
API Layer	Express, FastAPI
Auth	Login, JWT, Sessions
Database	MongoDB, PostgreSQL
Cache	Redis
Cloud	AWS, Vercel
DevOps	Docker, CI/CD
Security	HTTPS, hashing
Monitoring	Logs, uptime

Angela introduces the **core stack** used by startups.

2 Node.js (Runtime)

Node.js allows JS to:

- Run on servers
- Handle thousands of users
- Be async & fast

Why companies love Node:

- Same language frontend + backend

- Huge library ecosystem
 - Perfect for APIs & startups
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3

Express.js (Backend Framework)

Express is your **API engine**

It gives:

- Routing
- Middleware
- Request handling
- Response handling

Without Express → low-level pain

With Express → production backend



4

npm (Package Manager)

npm is how backend devs:

- Install libraries
- Manage versions
- Build fast

Every backend project = `package.json`

This is your **backend dependency map**

**5**

MongoDB (Database)

MongoDB stores data as **documents** (JSON)

Example:

```
{  
  "email": "shaurya@gmail.com",  
  "password": "hashed123"  
}
```

Why Angela teaches MongoDB:

- Easy
 - Flexible
 - Perfect for web apps
 - Used by startups
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**6**

Mongoose (MongoDB Tool)

Mongoose gives:

- Schemas
- Validation
- Models
- Queries

Without Mongoose → messy DB

With Mongoose → clean data

**7**

bcrypt (Password Security)

Used for:

- Hashing passwords
- Preventing hacks

Passwords are never stored directly
They are:

password → hash → stored

Passport.js (Authentication)

Passport handles:

- Login
- Signup
- Google login
- Sessions
- OAuth

Used by:

- Facebook
 - Google
 - GitHub login systems
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Body-Parser

Reads form data from requests:

`req.body.email`

`req.body.password`

Without it, backend can't read forms.

10 **Postman**

Postman is used to:

- Test APIs
- Send GET, POST, PUT, DELETE
- Debug backend

Backend engineers use Postman daily.

11 **Git + GitHub**

Used for:

- Version control
- Collaboration
- CI/CD

Recruiters check your GitHub
Your backend must live there.

12 **Deployment Tools**

Angela uses:

- Localhost → Cloud

Industry uses:

- Render
- Railway
- Vercel
- AWS
- DigitalOcean

Deployment makes:

`localhost:3000` → `yourapp.com`

13 Docker (Industry)

Docker packages:

- Backend
- DB
- Environment

So your app runs **anywhere**

This is how startups scale.

14 Logging & Monitoring

Real backend has:

- Error logs
- Crash tracking
- Request logs

Tools:

- Winston
 - Sentry
 - Datadog
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Security Stack

Backend uses:

- HTTPS
- JWT
- CORS
- Rate limiting
- Input validation

This stops:

- Hackers
- Data leaks
- Fake requests